WELCOME TO MODULE 6

ADVANCE SYSTEM ADMINISTRATION

What we will learn…..

Chapter I- Role vs Features

Chapter II- Adding Roles And Features

Chapter III- What is domain controller

Chapter IV- Domain Controller and Active Directory

Chapter V- Active Directory Prerequisites

Chapter VI- What is DNS

Chapter VII- Active Directory Installation

Chapter VIII- Active Directory Users and Computers

Chapter IX- Active Directory User Account Management

Chapter x- Installing Windows client

Chapter XI- Joining the domain from windows 7 & 10

Chapter XI- Active Directory “Administrative Center”

Chapter XII- Active Directory “Domain and Trust”

Chapter XIII- Active Directory “ Module for Windows Powershell”

Chapter XIV- Active Directory “Site and Services”

Chapter XV- Active Directory Group Policy Management

Chapter XVI- DNS Administration

Chapter XVII- WebServer (IIS) Installation

Chapter I- ROLES VS FEATURES

- Roles

A server role is a set of software programs that, when they are installed and properly configured, lets a computer perform a specific function for multiples users or other computers within a network.

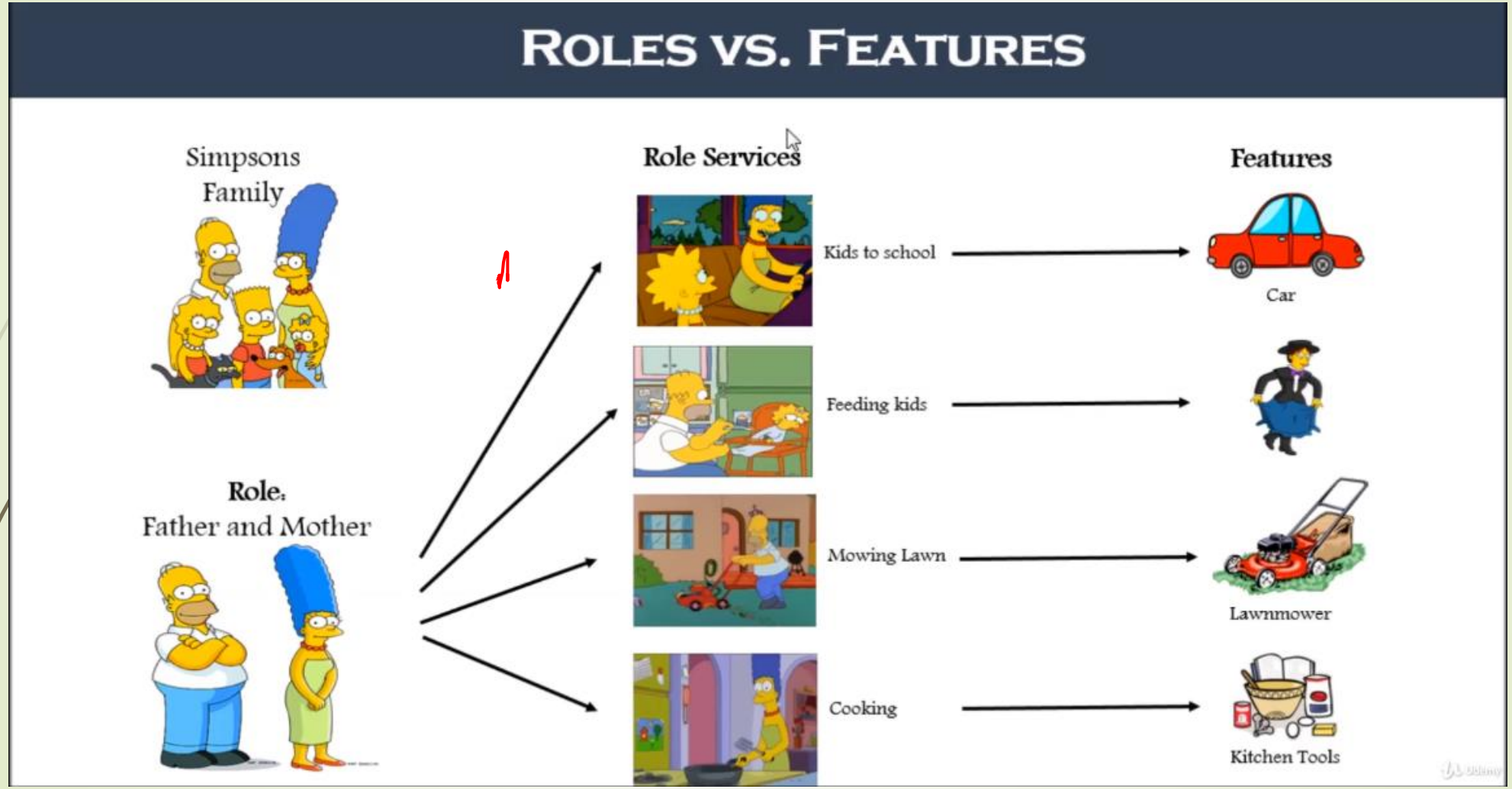
- Roles services

Are software programs that provide the functionality of a role. When you install a role, you can choose which services the role provides for other users and computers in your enterprise.

- Features

Are software programs that, although they are not directly part of roles or augment the functionality of one or more role, or improve the functionality of the server, regardless of which roles are installed.

Roles vs Features

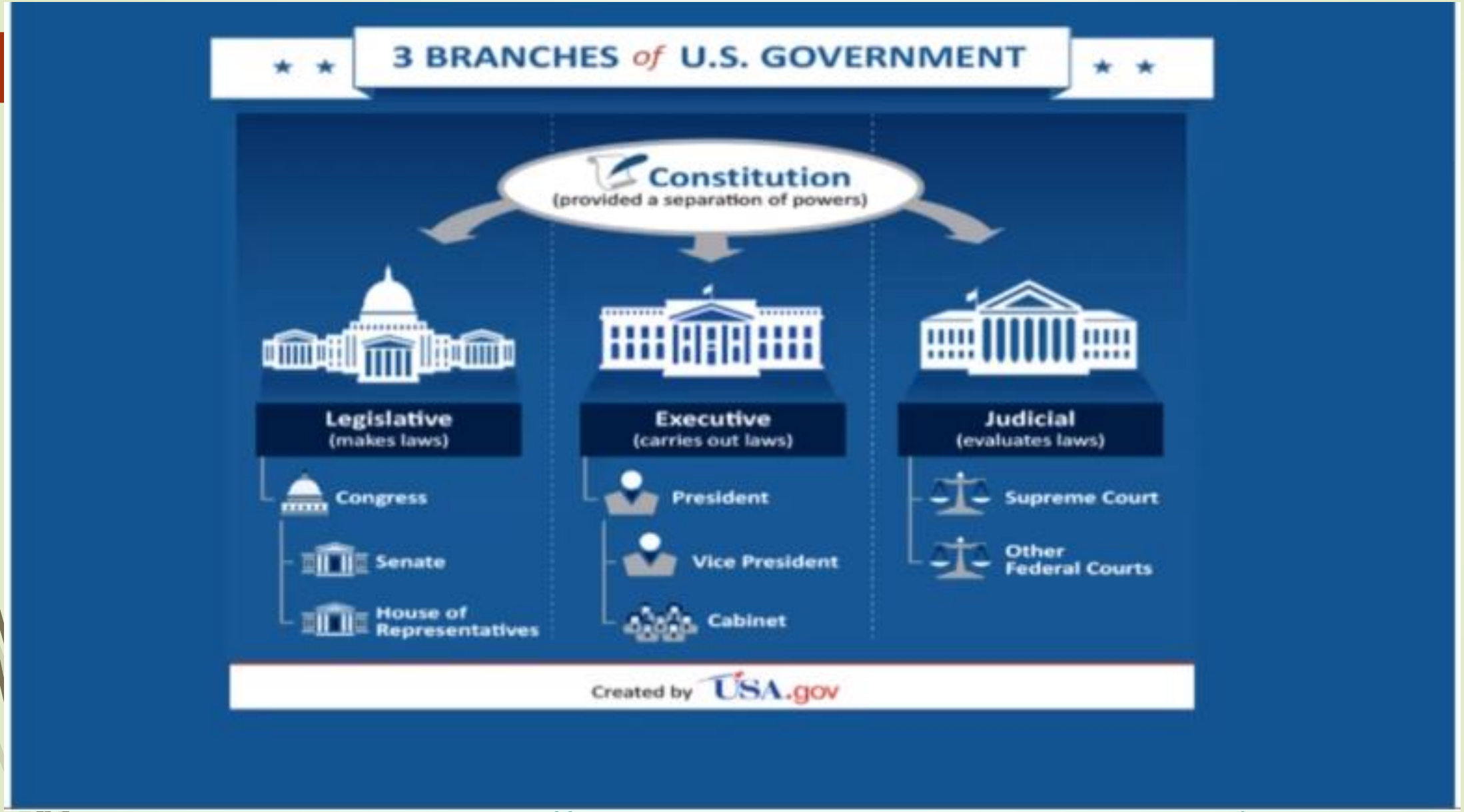


Chapter II- Adding Roles and Features

- When adding a role it may not requires service role or features.
- Service Roles are directly associated with roles and can't be installed without roles.
- When adding features it may or may not require service roles.
- When adding features it does not require roles.

Chapter III- Domain Controller

- Domain is an area of territory owned or controlled by a ruler or government
- Kingdom of resources
- Domain Controller Controls major group services such as:
  - User Activation (AD)
  - DNS
  - Print Services
  - Files System Separation
  - Windows Time Service
  - DHCP


3 BRANCHES of U.S. GOVERNMENT
Constitution
(provided a separation of powers)
Legislative
(makes laws)
Congress
Senate
House of Representatives
Executive
(carries out laws)
President
Vice President
Cabinet
Judicial
(evaluates laws)
Supreme Court
Other Federal Courts
Created by USA.gov

VALERY N.

Chapter IV- Domain Controller and Active directory

- Domain Controller (DC) is the main service that serves Active directory (AD) Service.
- Active directory is a directory service which allows user to authenticate logins, login to the clients.

1- Domain Controller Installation

- Domain Controller cannot be installed without a role
- e.g. you can open a company but it is useless if it does not provide any products or services.

- You will need a role service to install domain controller

2- WHAT IS ACTIVE DIRECTORY

- Active Directory is one of the primary service role of a domain controller
- It stores information about objects on the network and make this information available to users and network administrators. AD DS uses domain controllers to give network users access to permitted resources anywhere on the network through a single logon process.

Chapter V- Active Directory Prerequisites

- DNS
- Static IP ADDRESS
- DOMAIN CONTROLLER= mylabdc.local
- NETBIOS DOMAIN NAME= lab

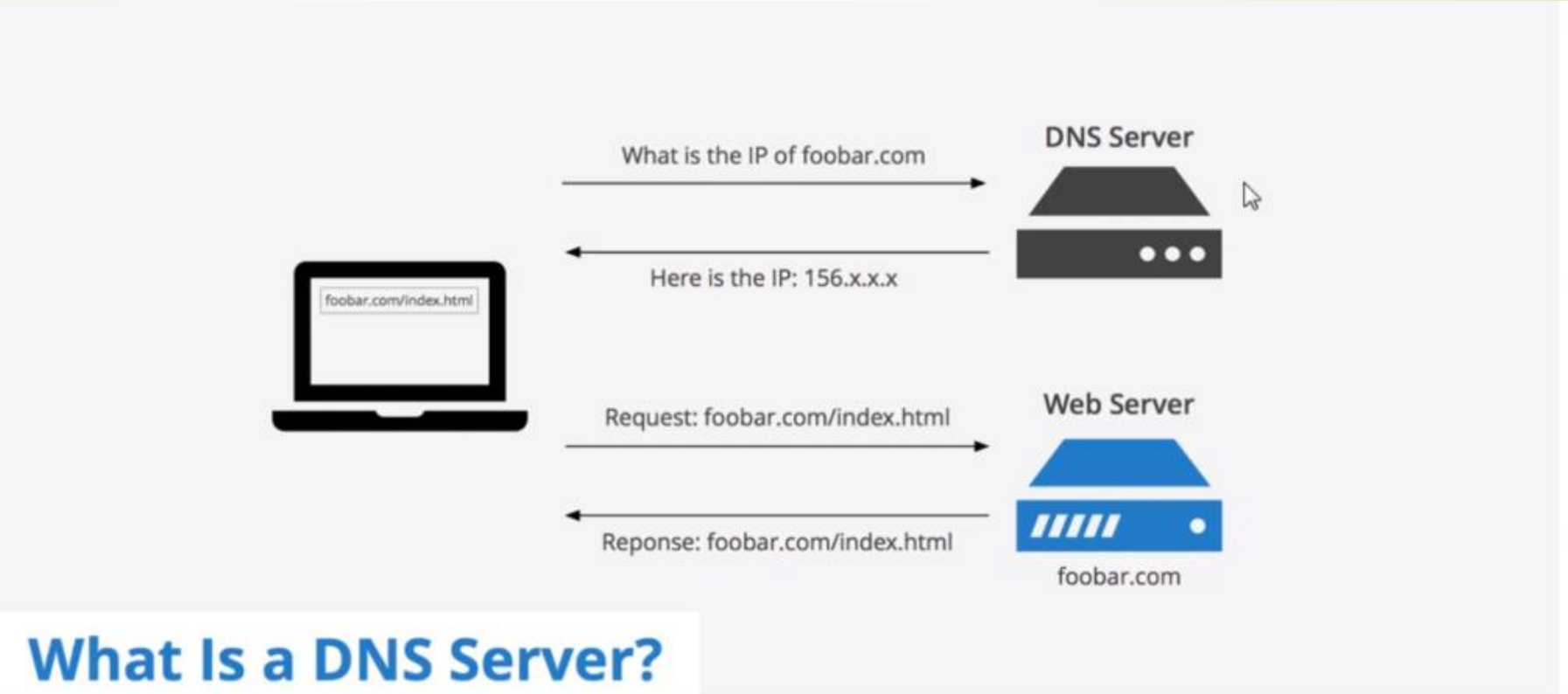
Chapter VI- What is DNS

- Domain Name Service
- Domain Name system or Server

- DNS is actually resolved the HOSTNAME to IP ADDRESS
- DNS also resolves HOSTNAME to IP ADDRESS
- It also resolves HOSTNAME to HOSTNAME (Alias or CNAME)

Cache memory is a chip-based computer component that makes retrieving data from the computer's memory more efficient. It acts as a temporary storage area ...

DNS


What is the IP of foobar.com
DNS Server
Here is the IP: 156.x.x.x
foobar.com/index.html
Request: foobar.com/index.html
Web Server
Reponse: foobar.com/index.html
foobar.com
What Is a DNS Server?

Chapter VII- Active Directory Installation

- DNS
- STATIC IP ADDRESS= 10.0.0.25
- DOMAIN CONTROLLER NAME= mylabdc.local
- NETBIOS Domaine name= lab
- **NSLookup** is a primary diagnostic tool for debugging a Domain Name System (DNS) architecture. You can use it to display any resource record on ...

What is the purpose of additional domain controller?

- Additional AD domain controller is **used to balance the load among existing domain controllers**. It also provides fault-tolerance that in case primary AD DC is down, additional AD DC can be used for authentications without any business discontinuity.
- If your one Domain Controller goes down users will be unable to log in to the domain, access resources in the domain, won't have access to their Exchange mailbox, etc…
- Additional AD domain controller is **used to balance the load among existing domain controllers**. It also provides fault-tolerance that in case primary AD DC is down, additional AD DC can be used for authentications without any business discontinuity.

Chapter VIII- Active Directory Users and Computers

- Organization unit

An organizational unit (OU) is a subdivision within an Active Directory into which you can place users, groups, computers and others organizational units.

- Objects

Are like Users, Groups, Contacts, Shared Folder, Printers etc…..

- Attributes

Objects have attributes that define and describe them. Form example, the attributes of a user object might include the user's name, e-mail address, and phone number.

Active Directory "Additional Information"

- Forest in AD

Forest is used to define an assembly of AD domains that share a single schema for the AD. All DC's in the forest share this schema and is replicated in a hierarchical fashion among them.

- AD runs on LDAP(lightweight Directory Access Protocol) protocol.

A lot of people mistake LDAP as an active directory. Remember that LDAP is a protocol uses for directory services. There are different directory services e.g: Active directory on windows and OPENLDAP on Linux.

- SYSVOL

The sysVOL keeps the server's copy of the domain's public files. The contents such as users, group policy, etc. of the sysVOL folder are replicated to all domain controllers in the domain.

- AD Database

AD database is saved into in C:\WINDOWS\NTDS

- Component of AD includes

✓ Logical structure, Trees, Forest and Domains or OU

✓ Physical structure, Domain Controller and Sites.

Chapter VIII- Active Directory Users and Computers

I. AD User Account Management

Managing Active Directory is one of the main responsibility of system administrator.

What we will learn…….

- Creating Organization Unit
  - HR, IT, FINANCE etc.

An organizational unit (OU) is a subdivision within an Active Directory into which you can place users, groups, computers, and other organizational units. You can create organizational units to mirror your organization's functional or business structure.

- Settings Sub-OU
  - IT= System, Network, Developer, Devops Etc.
  - FINANCE= Accounting, Taxation etc.

You can create a sub OU within the main OU.

We can create group within Sub OU e.g. Systems= Windows, Linux, VMWare.

Create a User e.g. Windows= James, Sean etc.

If you are working for a multi national company and you have office everywhere (Europe, India, Africa) each location will have an OU and Domain Controller then all Domains controller will be under the same Forest.

Example = Seinfeld TV Show

- OU = Seinfeld
- Sub-OU = Main Cast, Parents, Friends etc.
- Group = Parents → Jerry Parents, George Parents
- Users = Jerry Parents → Morty, Helen
  George Parents → Estelle, Frank

Chapter IX- Active Directory User Account Management

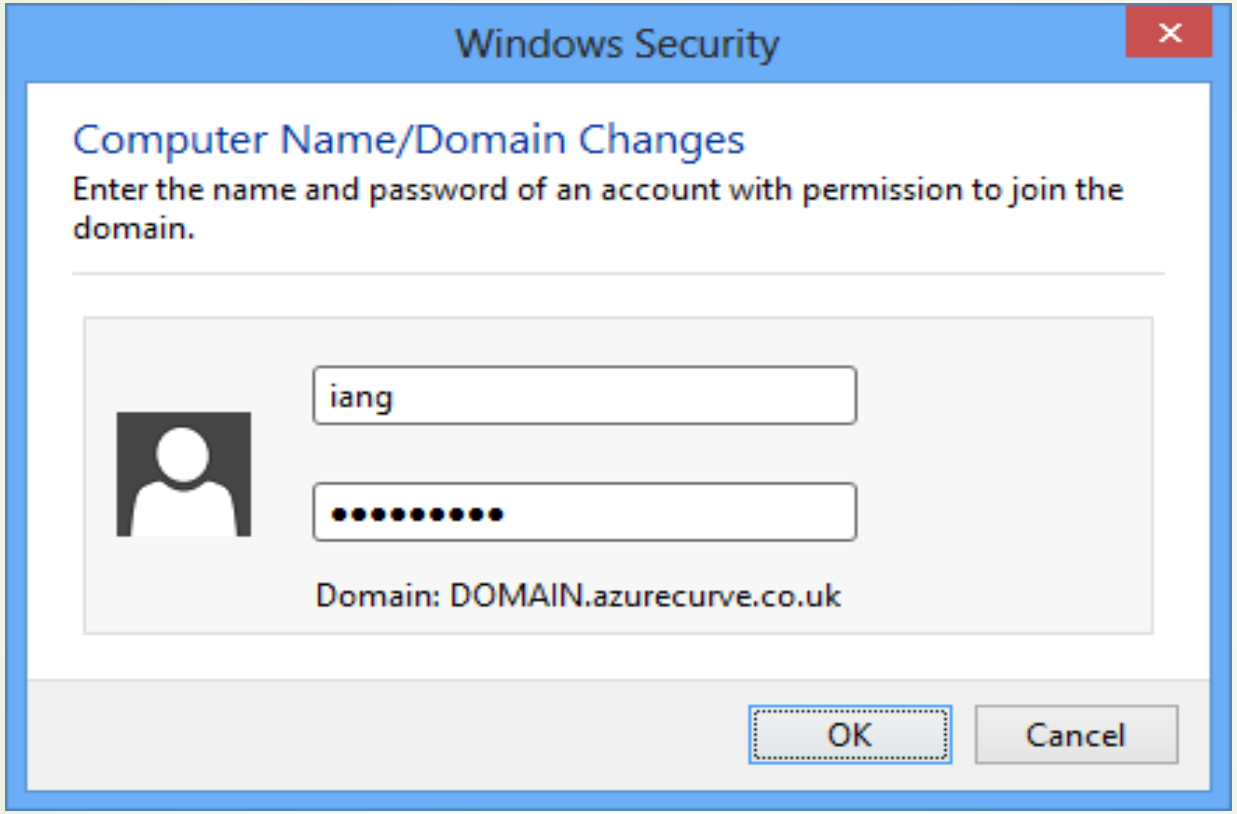
What will learn here……..

- How to create an OU(organization Unit)
- How to create a Sub OU
- How to create Users
- How to add Users into a Group

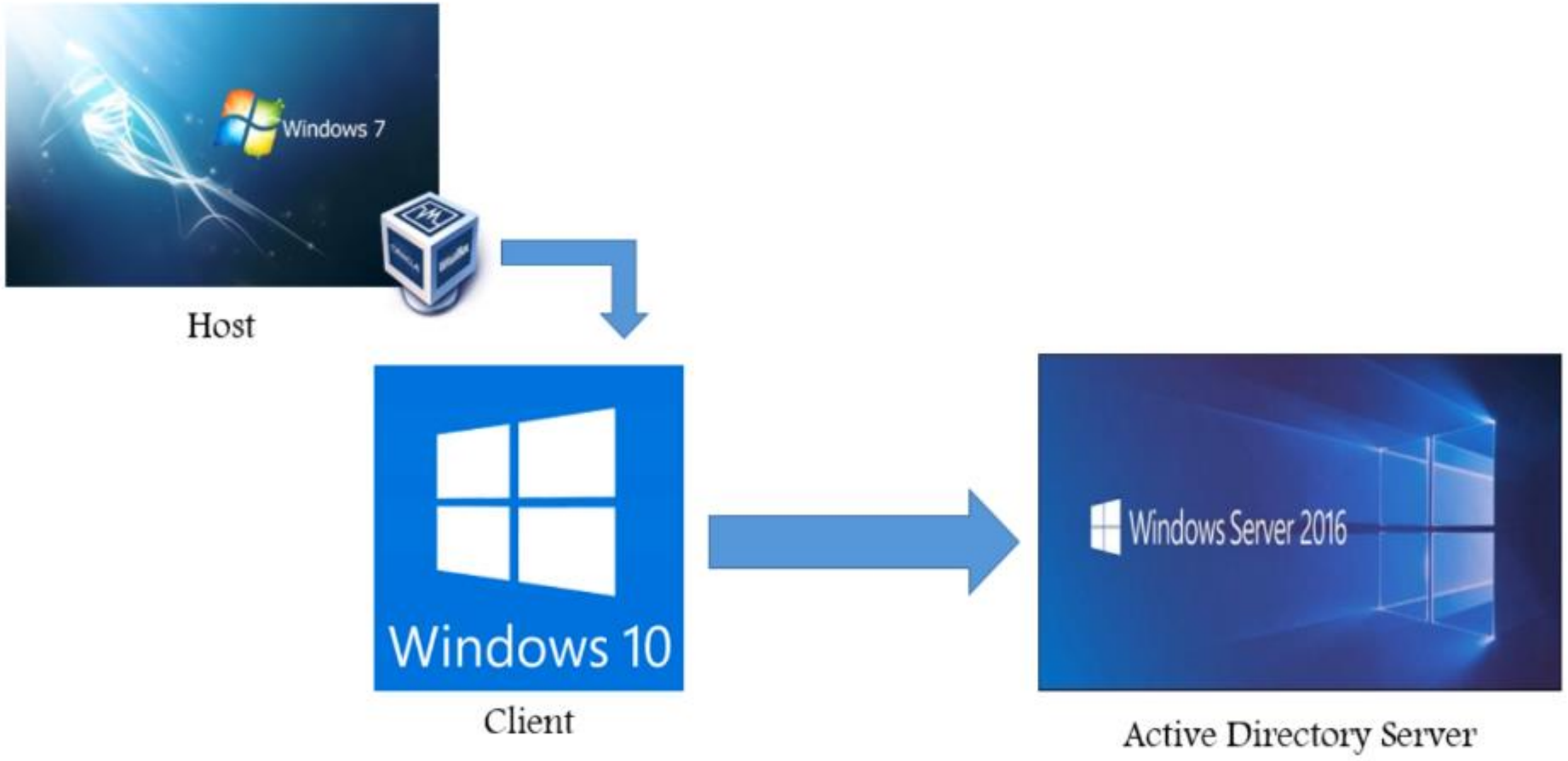
Chapter x- Installing Windows client

- Download windows 10 client in Microsoft website
- Create a virtual machine and install windows 10 on it.
- Change the hostname of the client machine.

Chapter XI- Joining the domain from windows 7 & 10



Chapter XI- Joining the domain from windows 7 & 10



PREREQUISITES FOR JOINING THE DOMAIN

- AD should be up and running with static IP
- Client can reach the active directory.
- A User account should exist on active directory.

Chapter XI- Active Directory "Administrative Center"

- **Active Directory Administrative Center**

  The Active Directory Administrative Center (ADAC) in Windows Server includes enhanced management experience features.

  These features ease the administrative burden for managing Active Directory Domain Services (AD DS)

  Some of the features are:

  - Search for OU, groups, users etc
  - Reset user password
  - Create objects.

Chapter XII- Active Directory "Domain and Trust"

- The active directory domains and trust console is a standard Microsoft management console (MMC) with the usual layout and elements. The left pane shows the domain list and the right pan shows objects, such as trust, associated with the selected domain.

Two of main features are:

- Manage Domain trust = There are several tasks you can perform with the console, including verifying or removing a trust and creating shortcut, realm, and external trusts. These trusts task will be explain later.
- Manage forest trust = you can accomplish several tasks related to forest trust, including creating a forest trust and managing routing for specific names suffixes

Chapter XIII- Active Directory " Module for Windows Powershell"

- The PowerShell cmdlets from the Active Directory module interact with the web service that is part of the domain controller with the ADDS role or ADMGS (AD management gateway service)

- You can install the Active Directory module for Windows PowerShell not only on the domain controller, but also on any Windows server or workstation

- Command to list all AD commands
  - `Get-Command -Module ActiveDirectory`

INRODUCTION TO POWERSHELL

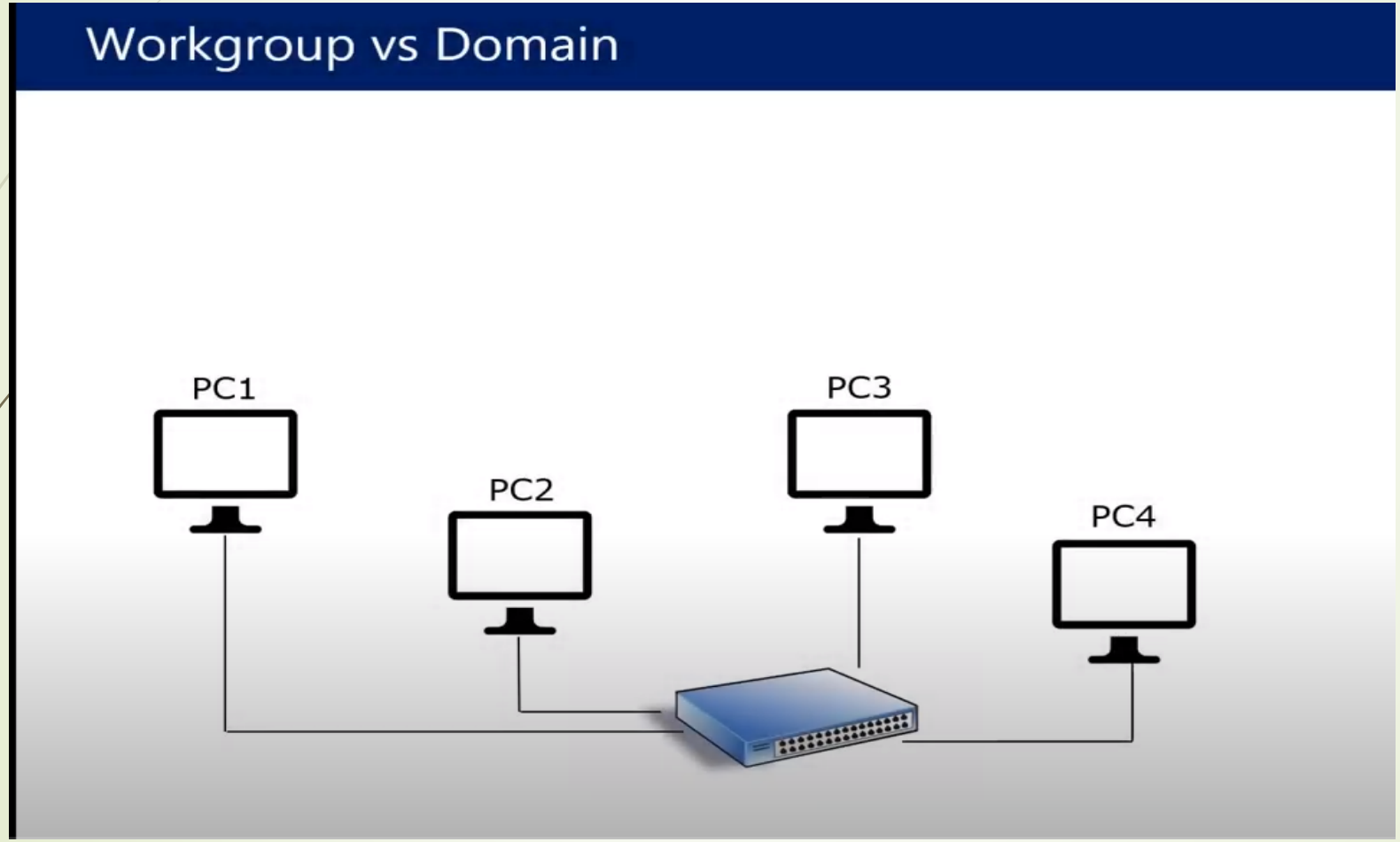
- $PSVersionTable: will display powershell version.
- Invoke-WebRequest: Download a file or app from the internet
- Get-commands will display all available commands
- New-Aduser to create a new user
- Get-Hel for help
- Update-Help
- Get-Help Get-Process –online
- import-module ActiveDirectory

- Powershell is also defined as command line management and administration environment.

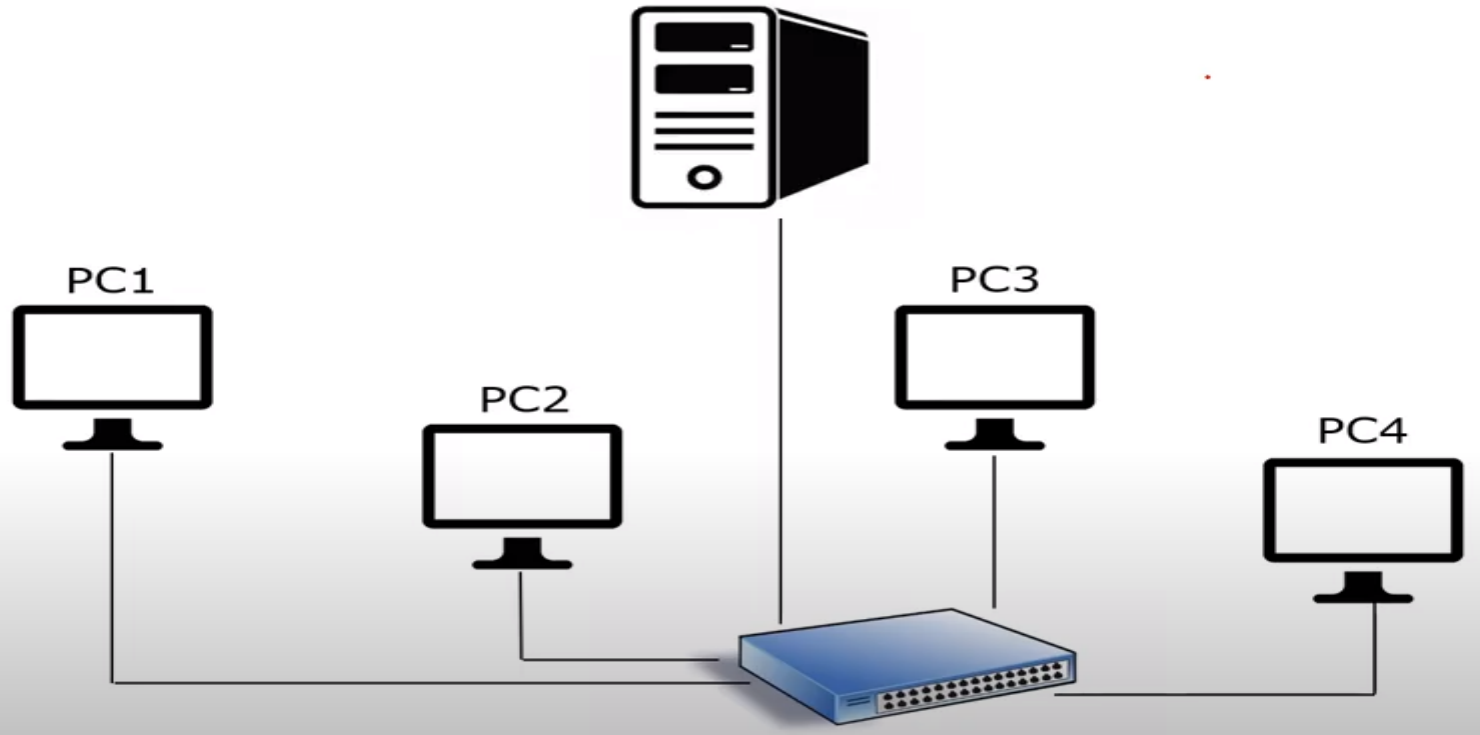
WSL

- Windows Subsystem for Linux is a compatibility layer for running Linux binary executables natively on Windows 10, Windows 11, and Windows Server 2019. In May 2019, WSL 2 was announced, introducing important changes such as a real Linux kernel, through a subset of Hyper-V feature.
- WSL **provides a layer for mapping Windows kernel system calls to Linux kernel system calls**. ... Instead of using a VM or container, WSL virtualizes a Linux kernel interface on top of the Windows kernel. This means that running WSL only requires a minimal amount of RAM.

WORKGROUP VS DOMAIN

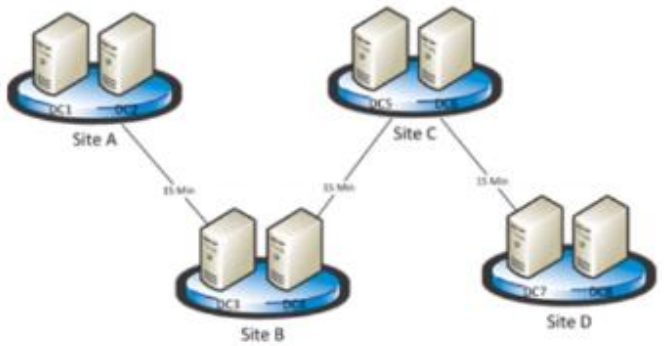


Workgroup vs Domain



Chapter XIV- Active Directory "Site and Services"

Sites are primarily used for replication and authentication optimization



- Subnets
  - Server in different subnets
- Inter-Site Transports
  - IP
    - Every new domain controller will create IP site link
    - You can setup site replication here
    - Site that is close by on the same network
    - High speed network to another office
  - SMTP
    - Slow or un-reliable connection

Why do you need to create site and services ?

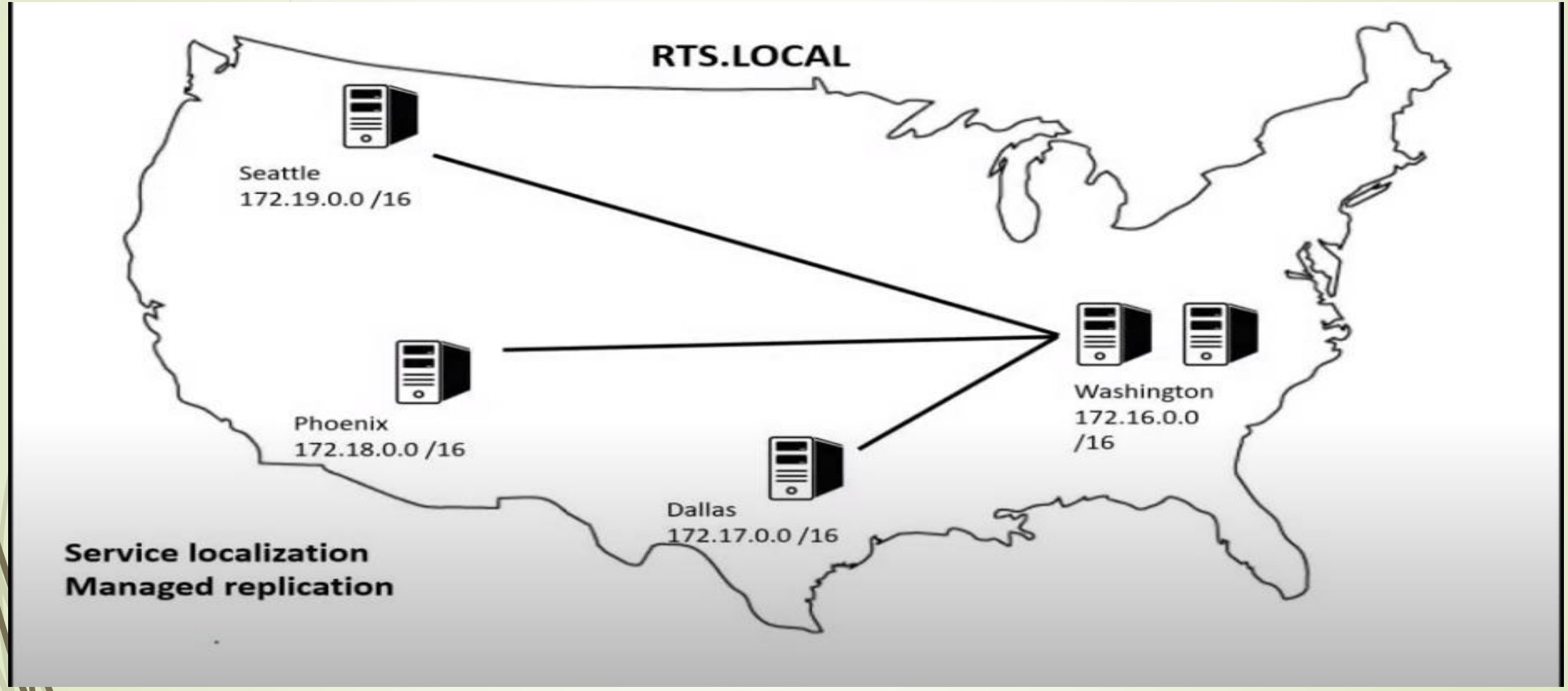
From a user point view which domain controller do I actually if I am a user in Seattle ?

If a user logon in Seattle office we what him to talk to a DC in Seattle.

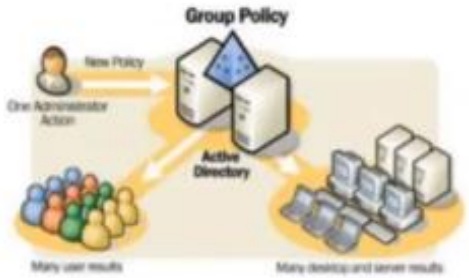
- Services Localization, you will always use local active directory first in your physical location before you cross a wired link and by pass a DC in your location and go to Washington or Phoenix to authenticate

- Managed Replication, that means every domain controller within the same active directory domain will be a replica of each other.

PRATICE SCENARIO



Chapter XV- Active Directory Group Policy Management



- **Group Policy** allows a network administrator in charge of Microsoft's Active Directory to implement specific configurations for users and computers. Group Policy can also be used to define user, security and networking policies at the machine level

- **Local group policy**
  - gpedit.msc
- **Domain group policy**
  - Same as gpedit.msc but applied from DC

Chapter XVI- DNS Administration

DNS records are instructions that live in authoritative DNS servers and provide information about a domain including what IP address is associated with that domain and how to handle requests for that domain. These records consist of a series of text files written in what is known as DNS syntax. DNS syntax is just a string of characters used as commands that tell the DNS server what to do. All DNS records also have a 'TTL', which stands for time-to-live, and indicates how often a DNS server will refresh that record.

- Type of DNS records
  - **A record** (host ipv4) that's resolution for hostname to ip address.
  - **AAAA record** (host ipv6) ipv6 is an enhance version of ipsec
  - **PTR record** (reverse) it will create a resolution for ip address to hostname.
  - **CNAME record** (Alias) Forwards one domain or subdomain to another domain, does NOT provide an IP address.
  - **MX record** - Directs mail to an email server.

Chapter XVII- WebServer (IIS) Installation

- A web server is server software, or a system of one or more computers dedicated to running this software, that can satisfy client HTTP requests on the public World Wide Web or also on private LANs and WANs
- A **web server** delivers content for a website to a client that requests it, like a web browser. A **web server** communicates with a web browser using HTTP.

- Internet Information Services (IIS, formerly Internet Information Server) is an extensible web server created by Microsoft for use with the Windows NT family
- It takes second place in overall usage behind Apache HTTP Server on the internet. It will host websites, web applications and services needed by users or developers
- IIS supports protocols such as HTTP, HTTP/2, HTTPS, FTP, FTPS, SMTP and NNTP

We will learn…

- How to install IIS (webserver)
- Access the initial webpage
- How to access IIS manager
- Create a new webpage.

Chapter XVIII- SERVER CORE OPERATING SYSTEM

- Windows **Server Core** is a minimal installation option for the Windows **Server operating system** (**OS**) that has no GUI and only includes the components required to perform **server** roles and run applications.

- Has no local graphical user interface
- Approximately 6.5 GB disk space required
- Reduced attack surface
- Reduced Windows updates

Support most Windows roles and features including:

- Active Directory Domain Services (AD DS)
- DHCP Server
- DNS Server
- File Server
- Print Server
- Hyper-V Server
- Failover clustering
- BitLocker Drive Encryption

Chapter XIX- NANO SERVER

Nano Server Update

- Nano server is no longer supported by Microsoft and not included in current or future versions of Windows Server 2016. However, some environments may have Nano servers that they have not yet migrated to other platforms.
- This course will cover Nano Server so students are familiar with the concept and for informational purposes only.
- Do not build your own Nano Server